Project Proposal

Human factors research has been a crucial part of reducing accidents in high pressure and high stakes environments. From aviation to nuclear plants, understanding when things go wrong and why informs researchers on how to find preventative measures. In class, one of our discussions argued that technology and data are not neutral and are filled with biases of the people collecting, analyzing, and interpreting data. In a field that relies heavily on objectivity for human factors analysis, the inability to have neutral data can influence how human factors researchers interpret previous accidents and implement solutions.

One Human Reliability Analysis (HRA) method was a second generation method developed by Erik Hollnagel called the Contextual Control Model (COCOM) of Cognitive Reliability and Error Analysis Method (CREAM). CREAM's goal is to establish "links between the operator's cognitive grade and human error probability," otherwise known as HEPs, and give "the probability of occurrence of the operator's behavior under different control modes" (Pan et al., 2017). COCOM was theorized and built off of CREAM because "human performance was discovered to not be an isolated random action but to be dependent on and controlled by the environment in which the task was completed" (Pan et al., 2017).

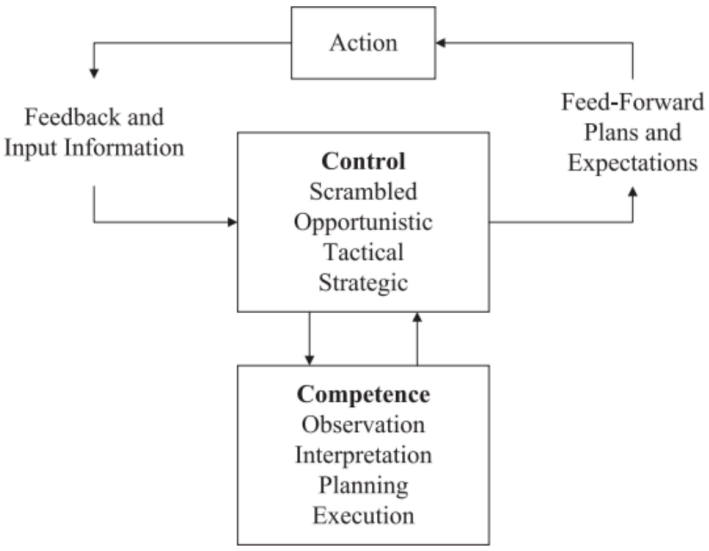


Figure 7. Overview of contextual control model (adapted from Hollnagel[44])

Fig 1. Screenshot of COCOM model (Pan et al., 2017).

The Multi-attribute Technological Accidents Dataset (MATA-D) utilizes COCOM to analyze the interactions between human, technology, and organizational factors and classify the

types of error observed in “238 major accidents from different industrial backgrounds” (Moura et al., 2016). Our goal for this project is to understand how biased MATA-D is and in what ways that bias is reflected in the way researchers use and interpret the dataset.

Research Questions:

We have 3 groups of research questions that have qualitative and quantitative components:

- How has the categorization of incidents changed over time? Have there been changes in the number or type of accident? Were the same accidents categorized differently during different years?
- Does this dataset put too much blame on the people instead of the system more than it should? If yes, how are these biases reflected on the AI model trained on this dataset?
- Is the incident adequately described within the categorizations? Are some errors more detrimental than others? Are some errors correlated with others?

description of at least one dataset, details of what you would expect to find, rough sketch of how you plan to analyze your data. You can think of what you would expect to find like your research hypotheses, but please keep in mind that your question does not have to be inferential in nature.

Dataset: Multi-Attribute Technological Accidents Dataset (MATA-D)

- How has the categorization of incidents changed over time?
    - Have there been changes in the number or type of accident?
    - Analyze by year and see if there is a higher prevalence of certain types of errors – proportion but also model occurrences by year
    - Check also the number of errors
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- Are some errors correlated with others?

The MATA-D dataset is a catalogue of 238 major accidents in high-technology industries between the years of 1953 and 2020. In MATA-D, each incident has the mistakes (marked by a 1) categorized under different classifications of error. Each tally has a comment with an explanation of the mistake and why it was categorized as where it is. Our first step to analyzing our data, aside from cleaning and formatting these tallied mistakes, will be to appropriately format the mistake comments for qualitative analysis. This allows us to view our dataset as two: one purely quantitative and the other purely qualitative.

Our first research question regarding the change in the categorization and frequency of

accidents over time can be answered using this dataset by comparing the reports from accidents that took place in different years and seeing what column of the table they were filed under. Since the data spans such a long time period, it is reasonable to expect that the ways in which the data was observed and filed will have changed over time, even if the authors categorized all the data in the same year.

This can also be connected to our second research question regarding whether the dataset puts too much of the blame for accidents on people rather than on institutional or systemic failures. This is something that also may have changed over time. For example, it is possible that an accident caused by a human misusing a piece of technology may have originally been categorized as a human-caused mistake, whereas a more modern example of a similar mistake may be filed as an institution problem due to the company providing insufficient training or improper technology.

Categorizational ambiguity such as the example described above could lead to editorializing and the introduction of bias into the dataset. Bad actors or imprudent researchers analyzing this dataset or creating models based on it will then reflect the bias.

This observation will be analyzed through the qualitative commented aspect of the dataset, where mistakes of the same kind being categorized differently may allude to our second research question.

Group Contract and Work Agreement

Your group contract and work agreement doesn't count for the 500-1000 word count of your proposal. This part will also only be graded on completion, not accuracy. The point of this is to prompt you to think about and write down your group work expectations.
You must discuss with your group expectations you have for one another and come to an agreement on an equitable division of labor. Please summarize this discussion and these expectations in one or two paragraphs after your proposal.
By division of labor I do not mean that you need to decide who will do what parts of the project at the beginning; I mean that you should come to an agreement about the amount and quality of work that is expected of each group member.

- Everyone should understand what other people are doing/other components they might not directly be contributing to
- Thorough commenting and recording of work for others to understand the work done
- Everyone should use the time spent on the project to get as much organized usable work done. This work does not need to be perfect but should contribute towards our goal of doing the group project well
- We should meet at least once a week to work on the project together

- If any one of us is unable to meet, do the work offline and communicate progress through text as to remain on schedule for deadlines
- Ensure that work missed due to conflicts/illness etc is made up for later as to ensure that each group member contributes sufficiently to the project

References:

Pan, X., Lin, Y., and He, C. (2017) A review of cognitive models in human reliability analysis. *Quality and Reliability Engineering International, 33*, 1299–1316. doi: 10.1002/qre.2111.

Moura, R., Beer, M., Patelli, E., Lewis, J., and Knoll, F. (2016) *Learning from major accidents to improve system design*. *Safety Science, 84*, 37–45. doi: 10.1016/j.ssci.2015.11.022.